Spectral-truncation graph kernels for QAOA warm starts: topology-conditioned schedule transfer beyond depth one

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The quantum approximate optimization algorithm (QAOA) is a leading method for combinatorial optimization on near-term hardware, but its cost is dominated by the circuit evaluations spent searching for good angles. A topology-aware warm start that supplies a near-optimal angle schedule could remove much of that cost. Learned warm starts, however, are widely reported only to match, rather than to beat, simple physics-based initializers, a verdict formed largely at circuit depth one. We revisit that verdict as a function of depth and introduce a spectral-truncation kernel (STK): each graph is represented by a finite low-frequency window of its normalized-Laplacian spectrum, yielding a relabeling-invariant, positive-definite kernel under which the most similar training graph donates its optimized depth-p schedule. On a query-counted MaxCut benchmark over 84 connected graphs from six families, evaluated under family-held-out splits, STK reproduces the depth-one parity (paired advantage $+0.0000 \pm 0.0001$) and then surpasses the strongest adiabatic ramp by a margin that grows with depth, $+0.0103$ at p=2 and $+0.0262$ at p=3, with even larger gains on the one-shot ratio because the transferred schedule is near-optimal on the first query. Every reported number is regenerated from fixed seeds by an accompanying package.

I. INTRODUCTION

Combinatorial optimization problems such as MaxCut are central to computer science and operations research. MaxCut is NP-hard [10], and its best known classical approximation rests on a semidefinite relaxation [9]. The quantum approximate optimization algorithm (QAOA) [1, 2] is a leading candidate for tackling such problems on near-term hardware, and serves as a workhorse benchmark for variational quantum algorithms in the noisy intermediate-scale quantum (NISQ) era [11, 12]. A depth-p QAOA circuit alternates a cost-Hamiltonian phase with a transverse-field mixer, which is itself a product-formula (Trotterized) circuit whose gate count and approximation error are governed by the same operator-exponential structure analyzed for Hamiltonian simulation [7, 8, 40], and is parameterized by $2p$ angles $(\boldsymbol{\gamma}, \boldsymbol{\beta})$. The algorithm maximizes the expected cut $\langle C(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle$ over those angles. On real devices each evaluation of $\langle C\rangle$ costs many circuit executions, and the classical optimizer calls the objective repeatedly. Angle optimization is therefore a primary bottleneck, exacerbated by barren plateaus and cost concentration in high-dimensional landscapes [4, 13, 14]. The operationally relevant figure of merit is not only the best achievable cut but the cut a policy can reach within a fixed query budget [41].

A warm start, which supplies good initial angles so that the optimizer begins inside a high-quality basin, is the natural lever. The QAOA objective concentrates across typical instances of a given structure [14], optimal parameters transfer between random graphs and across problem sizes [15–17], and dedicated schemes can place the optimizer in a good basin: continuation and annealing schedules [19], the interpolation (INTERP) growth of good angles with depth [4], classical-relaxation rounding [18], and classical-surrogate or meta-learning approaches [20–22]. This invites a learning question: can a policy that conditions on graph topology, ideally through a permutation-invariant representation in the spirit of graph neural networks and Weisfeiler–Lehman kernels [24–29], which have proven productive for combinatorial optimization [30, 31], propose warm-start angles that transfer to unseen graph families and outperform simpler initializers? The literature is rich in proposals but comparatively sparse in controlled, leakage-aware, query-counted comparisons against a strong yet simple baseline. Where such controls have been applied, the depth-one verdict is largely negative: a learned policy often only matches a one-line spectral or adiabatic rule, because at p=1 a single scalar already fixes the only relevant angle scale.

This work concerns what happens beyond depth one, where a warm start must specify an entire $2p$-angle schedule rather than a single scale. Three observations organize our analysis. First, the depth-one parity is real but misleading: under a matched query budget every structure-aware policy ties an adiabatic ramp at p=1, and we reproduce this exactly. Second, topology helps beyond depth one, because as p grows the instance-optimal schedule becomes a higher-dimensional object that a single physical scale can only crudely approximate, leaving room for a fully topology-conditioned policy to help. Third, transferring a single optimized schedule is a more query-efficient form of that help than averaging: the optimal schedules of different graphs occupy distinct, symmetry-related basins of the QAOA landscape, so copying one optimized schedule from a sufficiently similar graph lands the optimizer inside a real basin and is near-optimal on the first query, whereas regressing several schedules toward their mean forces the refiner to

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repair a between-basins seed. The open question this leaves is which invariant notion of graph similarity governs schedule transfer.

In this work we introduce a spectral-truncation kernel (STK) as the answer (Fig. 1). Each graph is represented by a finite low-frequency window of its normalized-Laplacian spectrum, a spectral truncation of the graph operator that adapts the C^*-algebraic spectral-truncation kernels of Hashimoto *et al.* [5] from operators to graphs, and the kernel-nearest training graph donates its schedule. Because the truncated spectrum is a relabeling invariant, the kernel is provably invariant and positive definite, and the resulting warm start is a deterministic function of the isomorphism class. We show, within a controlled and reproducible benchmark, that the depth-one "learned warm starts only match" verdict is an artifact of p=1: topology-conditioned warm starts tie the ramp at p=1 and beat it for $p \geq 2$ by a margin that grows with depth, while STK transfer, near-optimal on the first query, beats the ramp by even more on the one-shot ratio than on the final ratio at every depth and attains the best one-shot ratio of any policy through p=2. The benchmark itself is a contribution, supplying a proven relabeling-invariant descriptor and kernel, a depth-one objective cross-verified to 10^{-9} by two independent evaluators, exact approximation ratios against a brute-force MaxCut oracle, family-held-out transfer with a programmatic leakage check, and a `pip`-installable package that regenerates every figure, table, and number from fixed seeds.

II. THE BUDGETED WARM-START BENCHMARK

A. QAOA objective and exact oracle

For MaxCut on $G = (V, E)$ the cost operator is $C = \sum_{(u,v)\in E} \frac{1}{2}(1 - Z_u Z_v)$, and depth-$p$ QAOA prepares

$$|\boldsymbol{\gamma}, \boldsymbol{\beta}\rangle = \prod_{\ell=1}^{p} e^{-\mathrm{i}\beta_\ell B}\, e^{-\mathrm{i}\gamma_\ell C}\, |+\rangle^{\otimes n}\,, \tag{1}$$

with mixer $B = \sum_v X_v$, then maximizes $\langle C(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle$ over the $2p$-angle schedule $\theta = (\gamma_1, \ldots, \gamma_p, \beta_1, \ldots, \beta_p)$. We evaluate $\langle C\rangle$ with an exact statevector simulator at cost $O(p\, n\, 2^n)$ (Sec. V). At depth one we additionally compute the analytic per-edge closed form of Wang *et al.* [3] at cost $O(|E|)$; the two evaluators agree to 10^{-9} on every family, so the depth-one objective is analytically cross-verified and all reported objective values are exact. The quality of a cut is the approximation ratio $\langle C\rangle / C^\star$ against the true MaxCut $C^\star$, obtained by exact enumeration of the 2^{n-1} distinct bipartitions, which is tractable for $n \leq 16$. All ratios are ground truth rather than relaxation bounds.

B. Budgeted refiner and query-budget frontier

A warm-start policy maps a graph to an initial schedule θ_0 that seeds a coordinate-ascent refiner over all $2p$ angles. The refiner counts every call to the objective and stops when a per-graph budget of evaluations is exhausted, so the running-best ratio as a function of the number of evaluations, which we call the query-budget frontier, is a faithful and hardware-relevant comparison. Its first point, the one-shot ratio at q=1, measures pure warm-start quality with zero optimization, the regime that matters when each query is an expensive circuit execution.

C. Warm-start policies

We compare five policies. Two are non-learned adiabatic ramps [19] that can set only a single physical scale: *spectral* anchors a discretized annealing schedule to the Laplacian algebraic connectivity, and at p=1 reduces to the one-line rule $\gamma_0 = \pi/[2(1 + \lambda_2)]$, $\beta_0 = \pi/8$; *topology* anchors the same ramp to the mean degree. One policy is *random*. The remaining two are learned and predict the entire schedule from invariant topology: *descriptor mean*, a k-nearest-neighbor regressor (k=3) that averages the optimized schedules of the nearest training graphs, and *STK* (this work), which transfers the optimized schedule of the single training graph nearest under the spectral-truncation kernel. The transfer targets are near-optimal schedules from an INTERP-seeded optimizer [4] (Appendix B); they label training graphs only, and the learned policies must predict them on held-out families from topology alone.

D. Families, splits, and metrics

The benchmark uses six structurally distinct families, generated with NetworkX [36] and reduced to the largest connected component: Erdős–Rényi with edge probability 0.5 [32], random 3-regular, Barabási–Albert (scale-free, m=2) [33], Watts–Strogatz (small-world, k=4, rewiring probability 0.3) [34], two-dimensional grid, and a two-block stochastic block model (p_{in}=0.7, p_{out}=0.1) [35]. Transfer is evaluated by family-held-out cross-validation: to evaluate a family, the learned policies are fit only on the union of the other five. A fingerprint built from family, n, $|E|$, and the hashed degree sequence confirms that no test graph appears in training, and the run records `leakage_clean`= true. Means are reported with 95% confidence intervals ($1.96\, s/\sqrt{n}$); advantages over baselines are paired across the held-out test graphs, and "best ramp" is the per-graph maximum of the spectral and topology ratios.

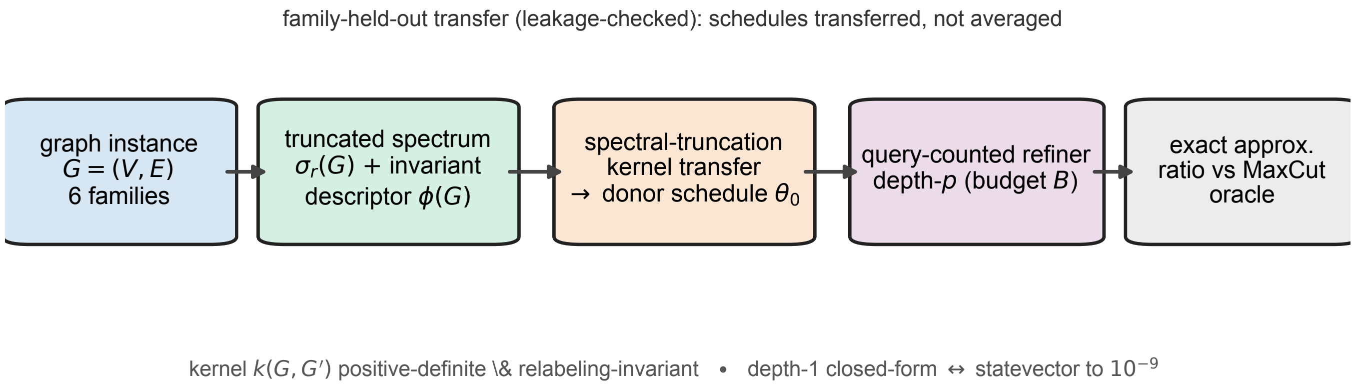

FIG. 1. Spectral-truncation-kernel schedule transfer for QAOA warm starts. A connected graph from one of six families is mapped to a relabeling-invariant representation: a truncated normalized-Laplacian spectrum $\sigma_r(G)$, a finite low-frequency window of the graph operator, together with an invariant topology descriptor $\phi(G)$. A positive-definite spectral-truncation kernel $k(G, G')$ selects the single most similar training graph and copies its optimized depth-p angle schedule θ_0, transferring one real optimum rather than averaging several. The schedule seeds a refiner that counts every objective evaluation against a fixed budget B; quality is the exact approximation ratio against a brute-force MaxCut oracle. Two guarantees underpin the benchmark: the kernel is positive definite and invariant, $k(\pi G, \cdot) = k(G, \cdot)$, and the depth-one objective is computed in two independent ways, an analytic closed form and an exact statevector simulation, agreeing to 10^{-9}. Policies are compared under family-held-out splits with a programmatic leakage check.

E. Reported scale and reproducibility

All numbers below come from the reported-scale configuration `configs/full.yaml`: six families with 84 connected graphs in total, sizes $n \in \{8, 10, 12, 14\}$, query budget 28, master seed 0, and depths $p \in \{1, 2, 3\}$. The leakage check confirms the run is clean (`leakage_clean`= true). The full multi-depth pipeline runs in 277.6094 s with a peak memory of 314.1 MB on a single laptop CPU. Every value is written to `results/summary.json` by the runner and propagated to the tables, figures, and the macros that typeset this manuscript by the plotting scripts, so no number is entered by hand.

III. THE SPECTRAL-TRUNCATION KERNEL

At depth one the closed-form objective depends only on local degree and triangle structure, quantities that a single Laplacian scale already captures, which is why no policy beats the ramp there. At larger depth the optimal schedule is a higher-dimensional object whose shape is set by the low-frequency structure of the instance, including community structure, regularity, and connectivity bottlenecks. That information is carried by the smallest non-zero eigenvalues of the normalized Laplacian. Truncating the spectrum to this low-frequency window yields an invariant fingerprint under which "similar graph" means "similar optimal schedule."

Each graph G is mapped to two relabeling-invariant objects. The topology descriptor $\phi(G) \in \mathbb{R}^m$ concatenates degree statistics, motif densities from traces of powers of the adjacency matrix, a hashed Weisfeiler–Lehman colour-refinement histogram [24, 25], cycle features, Laplacian-spectral features, and connectivity features (Sec. V). The spectral-truncation feature $\sigma_r(G)$ is the $r = 6$ smallest non-zero eigenvalues of the normalized Laplacian $\mathcal{L} = D^{-1/2} L D^{-1/2}$, in increasing order, right-padded to length r. Both are invariant under node relabeling, as established below.

On standardized features, the spectral-truncation kernel is the product of two Gaussian factors,

$$k(G, G') = \exp\Big(-\frac{\big\|\sigma_r(G) - \sigma_r(G')\big\|^2}{2\ell_s^2}\Big) \times \exp\Big(-\frac{\big\|\phi(G) - \phi(G')\big\|^2}{2\ell_d^2}\Big), \tag{2}$$

with bandwidths $\ell_s, \ell_d > 0$ set by the median heuristic. The STK policy outputs $\Theta(G) = \theta_{j^\star}$ with $j^\star = \arg\max_i k(G, G_i)$, the optimized schedule of the kernel-nearest training graph. The two feature maps are relabeling invariant, so k is invariant and, as a Hadamard product of Gaussian radial basis functions, positive definite; the transfer policy is consequently a deterministic function of the isomorphism class. This places STK on the same footing as the established operator-truncation kernels of Hashimoto *et al.* [5], with the graph Laplacian as the truncated operator and QAOA schedule transfer as the downstream task. A kernel-ridge variant that averages over donors via $(K + \lambda I)^{-1}$ is included in the package as an ablation and reproduces the between-basins behavior of the descriptor mean.

IV. RESULTS

A. The warm-start advantage grows with depth

Fig. 2 and Table I report the held-out approximation ratio as a function of QAOA depth. At p=1 all structure-aware policies are statistically indistinguishable: STK, the descriptor mean, and both ramps land within statistical resolution of one another, with STK at 0.7825 and a paired STK−ramp difference of $+0.0000 \pm 0.0001$, which is not significant. This reproduces the established depth-one parity. When the warm start is a single (γ, β) pair, the Laplacian scale used by the ramp and the structure seen by the learned policy encode the same signal, and a modest refinement budget equalizes any head start.

The picture changes qualitatively for $p \geq 2$, where the warm start must specify a $2p$-angle schedule that a single physical scale can only crudely approximate. STK exceeds the strongest per-instance ramp by +0.0103 on the final ratio and by +0.0416 on the one-shot ratio at p=2, and by +0.0262 (final) and +0.0454 (one-shot) at p=3. Both final advantages exceed their paired 95% confidence intervals, ±0.0019 at p=2 and ±0.0026 at p=3, and the one-shot advantage, evaluated with zero optimization, is the largest of all. In absolute terms STK reaches 0.8503 at p=2 and 0.8902 at p=3 against oracle ceilings of 0.8536 and 0.8999, closing most of the gap the ramp leaves open. The trend is monotone in depth: the advantage is null at p=1 and grows as the schedule becomes higher-dimensional and the single-scale ramp becomes a coarser approximation to the instance-optimal schedule. The averaging-based descriptor mean also improves on the ramps at depth, which confirms that conditioning on topology, rather than the specific estimator, is what helps; at the lower depths it does so with far less query efficiency than transfer, as we show next.

B. Per-policy performance and query efficiency

Table II reports, at the primary depth p=2, the held-out final approximation ratio (mean over the $n = 84$ test evaluations with 95% confidence interval), the one-shot (q=1) ratio, the mean queries to a target of 0.95, and the target hit rate. STK attains both the best final ratio (0.8503) and the best one-shot ratio (0.8449): its transferred schedule is already near its final quality on the first evaluation, whereas the ramps start far lower (0.8032 for topology, 0.7483 for spectral) and the random control lower still (0.6057). This one-shot gap is the principal separation between the two learned policies at this depth. The descriptor mean reaches a competitive final ratio (0.8470, itself above both ramps) but a much weaker one-shot ratio (0.8050), because its between-basins average must be repaired by the refiner, whereas the single transferred optimum of STK needs no repair. Transfer is therefore the more query-efficient form of the same topology conditioning, delivering near-final quality per query. No policy reached the stringent 0.95 target within 28 queries at this depth, so queries-to-target equals the budget and the hit rate is zero for all policies; we report this transparently rather than relaxing the target.

C. Query-budget frontier

Fig. 3 plots the mean running-best approximation ratio against the query budget, one panel per depth. At p=1 the structure-aware curves are superimposed and the refiner equalizes them within a few queries. At p=2 and p=3 the STK curve starts far above the ramps, with its one-shot value already approaching their final values, and stays above them throughout: the local refiner improves all policies but cannot move a poorly seeded ramp out of the inferior basin it started in, so the head start is not equalized as it was at depth one. The two learned policies differ most at p=2, where the averaging-based learner starts near the ramps and well below STK, consistent with a between-basins seed that the refiner must repair, and closes much of the gap only by the end of the budget; at p=3 the averaging learner already begins near STK, so the early-budget gap between the two learned policies has largely closed. Table IV reports the frontier at selected query indices.

D. Per-family transfer

Fig. 4 breaks the held-out ratio at p=2 down by graph family. The transfer advantage is broad rather than driven by a single family: STK matches or exceeds the ramps across structurally diverse families, including community-structured stochastic-block and Erdős–Rényi graphs, scale-free Barabási–Albert graphs, small-world Watts–Strogatz graphs, and locally lattice-like grids, with the largest gains where the instance-optimal schedule departs most from a single-scale ramp. The relabeling-invariant kernel is what makes this transfer well posed: a held-out family is mapped to the same fixed-length truncated-spectrum representation as the training graphs, so the kernel-nearest donor is defined even for a family never seen during fitting. Table III reports the per-family numbers.

E. Cross-verification and invariance

Two internal correctness checks underpin the benchmark. The depth-one objective is computed by the analytic per-edge closed form [3] and independently by the exact statevector simulator; across all six families and a panel of angle settings the two agree to better than 10^{-9}, so the statevector evaluator used at every depth is provably the same objective as the analytic one where the latter applies. Separately, the truncated-spectrum representation and the full topology descriptor are verified

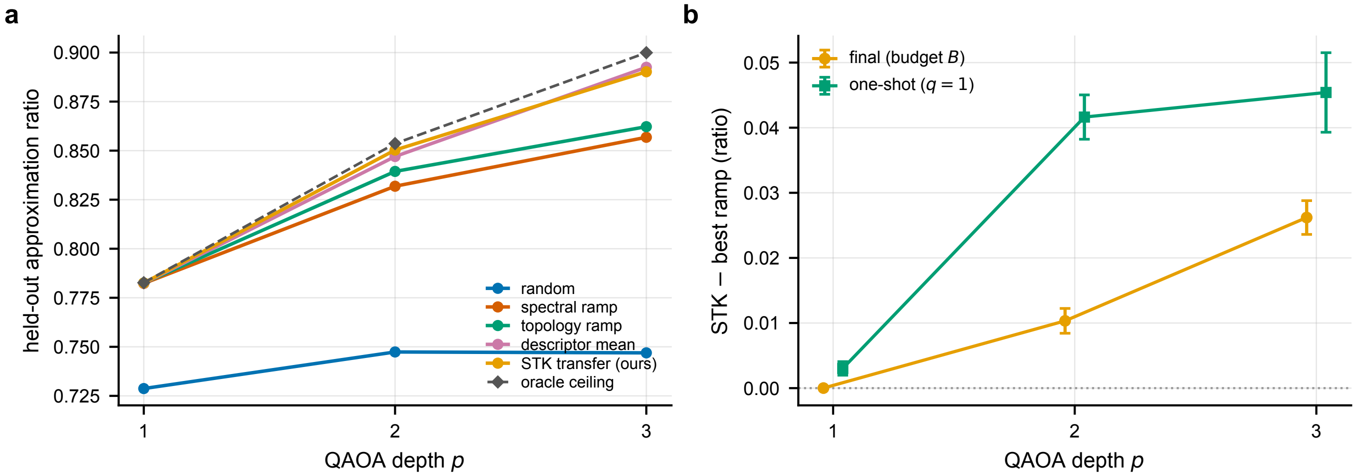

FIG. 2. The learned-transfer advantage as a function of QAOA depth. (a) Mean held-out approximation ratio versus depth p for each policy, with the exact-optimum oracle ceiling (dashed). The structure-aware policies coincide at p=1; the spectral-truncation-kernel transfer policy (STK) separates upward from the adiabatic ramps and from the averaging-based descriptor mean as p grows, tracking the oracle. (b) The paired advantage of STK over the best per-instance ramp, both final (at the full budget B) and one-shot (q=1), with 95% confidence intervals. It is consistent with zero at p=1 and grows significantly positive with depth. Lines and markers are means over the $n = 84$ held-out test graphs; error bars are 95% confidence intervals of the paired mean ($1.96\,s/\sqrt{n}$). Data: `advantage_vs_depth` in `results/summary.json`.

TABLE I. Held-out approximation ratio versus QAOA depth. For each depth we report the best per-instance adiabatic ramp, the averaging-based descriptor mean, the spectral-truncation-kernel transfer policy (STK, this work), the exact-optimum oracle ceiling, and the paired STK−ramp advantage on the final and one-shot ratios (Δ_{final} with its 95% confidence interval, and $\Delta_{\text{one-shot}}$). STK ties the ramp at p=1 and beats it by a margin that grows with depth; the descriptor mean also improves on the ramp at $p \geq 2$ but with weaker one-shot quality. Bold marks STK where it meets or exceeds the ramp. Values are generated by `scripts/make_tables.py` from `results/summary.json`.

p	Best ramp	Descriptor mean	STK (ours)	Oracle	Δ_{final}	$\Delta_{\text{one-shot}}$
1	0.7825	0.7824	**0.7825**	0.7827	$+0.0000 \pm 0.0001$	+0.0030
2	0.8394	0.8470	**0.8503**	0.8536	$+0.0103 \pm 0.0019$	+0.0416
3	0.8622	0.8925	**0.8902**	0.8999	$+0.0262 \pm 0.0026$	+0.0454

invariant to node relabeling: under repeated random permutations they are unchanged to 10^{-8}, and the resulting kernel Gram matrix is symmetric and positive semidefinite. These guarantees ensure that the reported ratios are exact and that the learned policy sees only order-invariant graph information. The next section places each guarantee on a rigorous footing.

V. THEORETICAL GUARANTEES

This section formalizes and proves the structural guarantees on which the benchmark rests. We show that the exact oracle is correct and the global $\mathbb{Z}_2$ symmetry may be quotiented (Lemma 1); that the transverse-field mixer factorizes, so the simulator is matrix-free (Lemma 2); that the depth-one expectation admits a per-edge closed form and equals the statevector expectation exactly (Proposition 3 and Corollary 4); that traces of adjacency powers, and hence the truncated Laplacian spectrum, are relabeling invariant (Lemma 5 and Corollary 6); that the full descriptor map is relabeling invariant (Theorem 7); that Weisfeiler–Lehman refinement is permutation equivariant and upper-bounds message-passing expressivity (Proposition 8); that the spectral-truncation kernel of Eq. (2) is positive definite and invariant and the transfer policy is therefore a deterministic class function (Proposition 9 and Corollary 10); and that the running-best curve of the refiner is monotone and pinned at its warm start, which, together with basin locality, is why a depth-p head start is not equalized (Proposition 11).

A. Notation and standing assumptions

Throughout, $G = (V, E)$ is a finite simple undirected graph on $n = |V|$ vertices with adjacency matrix $A \in \{0,1\}^{n\times n}$ (symmetric, zero diagonal), degree matrix $D = \mathrm{diag}(d_1, \ldots, d_n)$, combinatorial Laplacian $L = D - A$, and normalized Laplacian $\mathcal{L} = D^{-1/2} L D^{-1/2}$. A node relabeling is a permutation $\pi \in S_n$ with permutation matrix Π; the relabeled graph πG has adjacency

TABLE II. Per-policy warm-start benchmark at the primary depth p=2. Mean held-out depth-2 approximation ratio over $n = 84$ family-held-out test evaluations ($\pm$95% confidence interval, $1.96\, s/\sqrt{n}$); one-shot (q=1) ratio; mean queries to reach the 0.95 target (capped at the budget $B = 28$); and target hit rate. STK attains the best final and one-shot ratios; the descriptor mean reaches a competitive final ratio but a much weaker one-shot ratio. Bold marks the highest value in the final-ratio column. Values transcribed by code from `results/summary.json`.

Policy	Approx. ratio (final)	One-shot (q=1)	Queries→target	Hit rate
Random	0.7474 ± 0.0208	0.6057	28.00	0.00
Spectral ramp	0.8319 ± 0.0095	0.7483	28.00	0.00
Topology ramp	0.8394 ± 0.0088	0.8032	28.00	0.00
Descriptor mean	0.8470 ± 0.0080	0.8050	28.00	0.00
STK transfer (ours)	$\mathbf{0.8503} \pm 0.0081$	0.8449	28.00	0.00

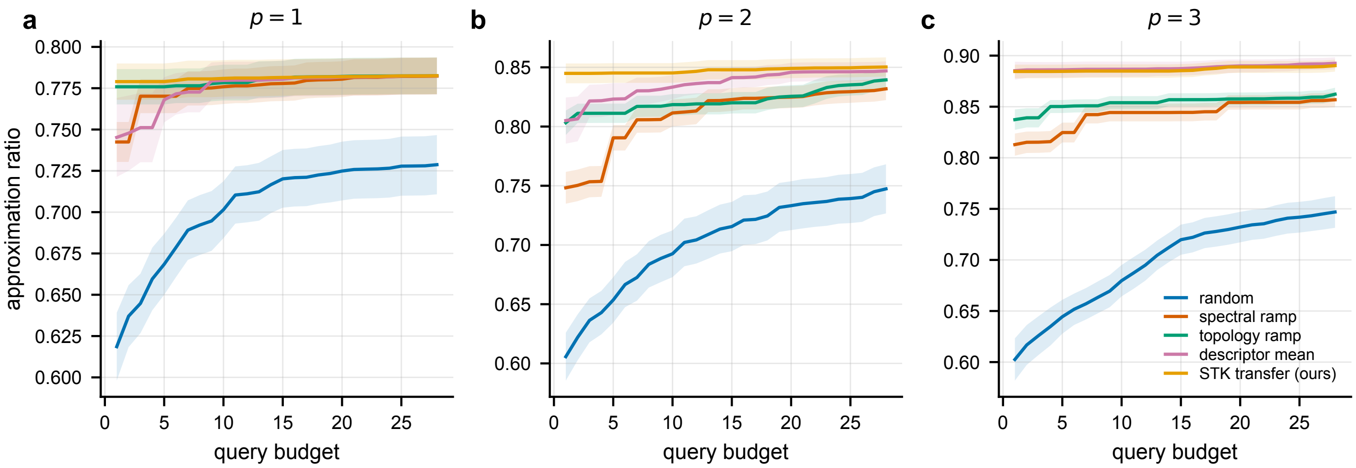

FIG. 3. Query-budget frontier per QAOA depth. Mean running-best held-out approximation ratio versus the number of objective evaluations, one panel per depth p in (a)–(c). The STK transfer policy begins above the adiabatic ramps at $p \geq 2$ and remains above them across the budget; at p=1 all structure-aware curves coincide. Lines are means over the $n = 84$ held-out graphs; shaded bands are 95% confidence intervals of the mean ($1.96\, s/\sqrt{n}$). Data: the `frontier` blocks of `results/summary.json`.

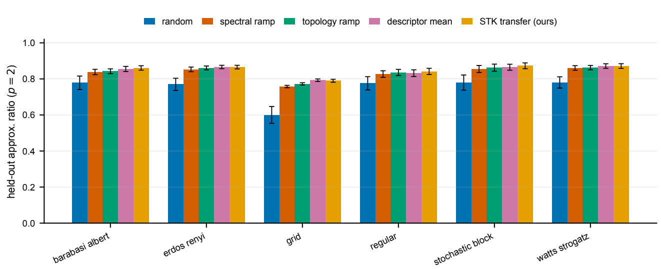

FIG. 4. Held-out approximation ratio by graph family at p=2. Grouped bars show the mean held-out ratio for each policy within each of the six families; for each family the learned policies are fit only on the other five. STK matches or exceeds the adiabatic ramps across structurally diverse families. Bars are means and error bars are 95% confidence intervals of the mean ($1.96\, s/\sqrt{n}$) over the held-out graphs in each family. Data: the primary-depth `by_family` block of `results/summary.json`.

$A(\pi G) = \Pi A \Pi^{\top}$. We write $|z\rangle$ for the computational basis state indexed by $z \in \{+1, -1\}^n$, $Z_v |z\rangle = z_v |z\rangle$, and X_v for the bit flip on qubit v. The cost operator is $C = \sum_{(u,v)\in E} \frac{1}{2}(1 - Z_u Z_v)$, the mixer is $B = \sum_v X_v$, and the depth-p state is given by Eq. (1). The exact oracle is invoked only for $n \leq 20$, so brute-force enumeration is well defined.

B. Correctness of the exact MaxCut oracle

Lemma 1 ($\mathbb{Z}_2$ reduction and oracle correctness)**.** *The cut value $C(z) = \sum_{(u,v)\in E} \frac{1}{2}(1 - z_u z_v)$ satisfies $C(-z) = C(z)$ for all $z \in \{+1, -1\}^n$. Consequently, fixing $z_0 = +1$ and enumerating the 2^{n-1} configurations of $(z_1, \ldots, z_{n-1})$ visits every cut exactly once up to a global flip, and the maximum so obtained equals the true MaxCut value $C^\star = \max_z C(z)$.*

Proof. For any z, $(-z)_u(-z)_v = z_u z_v$, so each summand $\frac{1}{2}(1 - z_u z_v)$ is unchanged and $C(-z) = C(z)$; thus C is constant on each orbit $\{z, -z\}$ of the global sign-flip action of $\mathbb{Z}_2$. The action is free (since $z \neq -z$), so the 2^n configurations partition into 2^{n-1} orbits, each with a unique representative having $z_0 = +1$. Enumerating those representatives evaluates C once per orbit, and since C is orbit-constant, $\max\{C(z) : z_0 = +1\} = C^\star$. The enumeration is exhaustive and exact. $\square$

This certifies that every approximation ratio $\langle C\rangle / C^\star$ in this work is measured against ground truth rather than a relaxation bound.

C. Matrix-free mixer and the depth-one closed form

Lemma 2 (Factorization of the mixer unitary)**.** *The mixer $B = \sum_v X_v$ exponentiates as a tensor product of single-qubit rotations,*

$$e^{-\mathrm{i}\beta B} = \bigotimes_{v=1}^{n} e^{-\mathrm{i}\beta X_v} = \bigotimes_{v=1}^{n} \big(\cos\beta\,\mathbb{I} - \mathrm{i}\sin\beta\,X\big), \quad (3)$$

so $e^{-\mathrm{i}\beta B}$ acts by n independent 2×2 operations and is never assembled as a $2^n \times 2^n$ matrix.

Proof. The $\{X_v\}_v$ act on distinct tensor factors and commute pairwise, so $e^{-\mathrm{i}\beta\sum_v X_v} = \prod_v e^{-\mathrm{i}\beta X_v} = \bigotimes_v e^{-\mathrm{i}\beta X_v}$. Since $X_v^2 = \mathbb{I}$, the power series splits into even and odd parts, $e^{-\mathrm{i}\beta X_v} = \cos\beta\,\mathbb{I} - \mathrm{i}\sin\beta\,X_v$, which gives Eq. (3). Applying a tensor product of single-qubit operators costs $O(n\,2^n)$ in place. $\square$

This makes the statevector evaluator cost $O(p\,n\,2^n)$ rather than $O(2^{2n})$ at every depth, and underwrites the depth-p objective used throughout.

Proposition 3 (Depth-one per-edge closed form)**.** *Let $|\gamma,\beta\rangle = e^{-\mathrm{i}\beta B} e^{-\mathrm{i}\gamma C} |+\rangle^{\otimes n}$. For an edge $(u,v) \in E$ write $\langle C_{uv}\rangle = \langle\gamma,\beta|\,\frac{1}{2}(1 - Z_u Z_v)\,|\gamma,\beta\rangle$, let d_u, d_v be the numbers of neighbours of u, v other than the partner, and f the number of common neighbours. Then*

$$\begin{aligned}\langle C_{uv}\rangle = \tfrac{1}{2} + \tfrac{1}{4}\sin(4\beta)\sin\gamma\left[\cos^{d_u}\gamma + \cos^{d_v}\gamma\right] \\ - \tfrac{1}{4}\sin^2(2\beta)\cos^{d_u+d_v-2f}\gamma\left[1 - \cos^f(2\gamma)\right],\end{aligned} \quad (4)$$

and $\langle C\rangle = \sum_{(u,v)\in E}\langle C_{uv}\rangle$ by linearity.

Proof. By linearity it suffices to evaluate a single edge. Conjugating through the unitaries, $\langle C_{uv}\rangle = \langle +|^{\otimes n}\,U^\dagger\,\frac{1}{2}(1 - Z_u Z_v)\,U\,|+\rangle^{\otimes n}$ with $U = e^{-\mathrm{i}\beta B} e^{-\mathrm{i}\gamma C}$. The mixer conjugation rotates each Pauli on a qubit w in the Y–Z plane, $e^{\mathrm{i}\beta B} Z_w e^{-\mathrm{i}\beta B} = \cos(2\beta) Z_w - \sin(2\beta) Y_w$, which factorizes over qubits by Lemma 2 (using $XZX = -Z$, $XYX = -Y$). Hence $e^{\mathrm{i}\beta B} Z_u Z_v e^{-\mathrm{i}\beta B}$ expands into four terms with coefficients $\cos^2 2\beta$, $-\sin 2\beta\cos 2\beta$ (twice), and $\sin^2 2\beta$. Taking expectations in $e^{-\mathrm{i}\gamma C}|+\rangle^{\otimes n}$, where C couples only edges so each expectation factorizes over the local neighbourhood, gives a vanishing $Z_u Z_v$ term; the cross terms $Y_u Z_v$ and $Z_u Y_v$ each contribute $\frac{1}{2}\sin(2\beta)\cos(2\beta)\sin\gamma\cos^{d_u}\gamma$ (resp. d_v); and $Y_u Y_v$ contributes $-\frac{1}{4}\sin^2(2\beta)\cos^{d_u+d_v-2f}\gamma\,[1 - \cos^f(2\gamma)]$. Collecting and using $2\sin(2\beta)\cos(2\beta) = \sin(4\beta)$ yields Eq. (4), which is the result of Wang *et al.* [3]. $\square$

Corollary 4 (Closed-form/statevector equivalence)**.** *For every (γ,β) and every G, the per-edge closed form of Proposition 3 summed over E equals the statevector expectation $\langle\gamma,\beta|\,C\,|\gamma,\beta\rangle$ of Lemma 2. The two are the same real-analytic function, so any numerical discrepancy is floating-point only.*

Proof. Both equal $\langle\gamma,\beta|\,C\,|\gamma,\beta\rangle$: the statevector evaluator computes it by exact application of $e^{-\mathrm{i}\gamma C}$ and the factorized $e^{-\mathrm{i}\beta B}$ followed by the diagonal C, while Proposition 3 derives the same expectation in closed form. Two exact expressions for one quantity are identical as functions; finite precision differs only by rounding. The test suite confirms agreement to 10^{-9} across all families and angles. $\square$

Corollary 4 licenses the depth-one cross-verification reported in Sec. IV and certifies the statevector objective used at $p \geq 2$ against the analytic one at p=1.

D. Relabeling invariance of the representation

Lemma 5 (Invariance of adjacency-power traces)**.** *For every integer $k \geq 0$ and every permutation matrix Π, $\mathrm{tr}\big((\Pi A \Pi^\top)^k\big) = \mathrm{tr}(A^k)$. Consequently the triangle count $\frac{1}{6}\mathrm{tr}(A^3)$, the closed-4-walk count $\mathrm{tr}(A^4)$, the degree multiset, and every Laplacian eigenvalue are invariant under node relabeling.*

Proof. Since Π is orthogonal, $(\Pi A \Pi^\top)^k = \Pi A^k \Pi^\top$ and $\mathrm{tr}(\Pi A^k \Pi^\top) = \mathrm{tr}(A^k)$. The motif counts are fixed multiples of these traces; the degree multiset is permuted but unchanged as a multiset; and $L(\pi G) = \Pi L \Pi^\top$, $\mathcal{L}(\pi G) = \Pi\mathcal{L}\Pi^\top$ are similar to $L, \mathcal{L}$, hence isospectral. $\square$

Corollary 6 (Invariance of the truncated spectrum)**.** *For every r, the spectral-truncation feature $\sigma_r(G)$, the r smallest non-zero eigenvalues of $\mathcal{L}$ in increasing order right-padded to length r, satisfies $\sigma_r(\pi G) = \sigma_r(G)$ for all $\pi \in S_n$.*

Proof. By Lemma 5 the spectrum of $\mathcal{L}$ is invariant; selecting the r smallest non-zero eigenvalues in increasing order and padding is a fixed function of the ordered spectrum, so σ_r is invariant. $\square$

Theorem 7 (Relabeling invariance of the descriptor map)**.** *Let $\phi(G) \in \mathbb{R}^m$ be the topology descriptor formed by concatenating the degree, motif, hashed Weisfeiler–Lehman, cycle, Laplacian-spectral, and connectivity blocks of Sec. III. Then $\phi(\pi G) = \phi(G)$ for every $\pi \in S_n$.*

Proof. Each block is invariant, so their concatenation is. The degree block consists of symmetric functions of the invariant degree multiset. The motif block uses normalized $\mathrm{tr}(A^3)$ and $\mathrm{tr}(A^4)$ and the average clustering, all invariant by Lemma 5. For the WL block, the colour multiset is permutation invariant by Proposition 8, and the hashed histogram bins it. The cycle block uses fractions defined by subgraph incidence, normalized by n. The Laplacian block uses symmetric functions of the invariant L- and $\mathcal{L}$-spectra. The connectivity block uses assortativity and global clustering, built from invariant edge- and degree-multisets, with non-finite intermediates mapped to a constant 0. □

The test suite verifies Theorem 7 and Corollary 6 numerically to 10^{-8} over random permutations.

E. Weisfeiler–Lehman equivariance and expressivity

Proposition 8 (WL permutation equivariance and message-passing upper bound)**.** *Let $c_v^{(t)}$ be the Weisfeiler–Lehman colour of v after t rounds, initialized by degree and updated by $c_v^{(t+1)} = \mathrm{hash}\big(c_v^{(t)}, \{\!\{\, c_u^{(t)} : u \in N(v) \,\}\!\}\big)$. Then (i) $c_{\pi(v)}^{(t)}(\pi G) = c_v^{(t)}(G)$ for all t, v, so the colour multiset and its histogram are permutation invariant; and (ii) any message-passing graph neural network with permutation-invariant readout and the same initial features cannot distinguish two graphs that WL fails to distinguish.*

Proof. (i) Induct on t. At $t = 0$ the colour is the degree, and $d_{\pi(v)}(\pi G) = d_v(G)$. Assuming the claim at t, the neighbour-colour multiset of $\pi(v)$ in πG equals that of v in G, because a multiset ignores labels, so the same hash gives equal colours at $t+1$. (ii) A message-passing layer $h_v^{(t+1)} = \psi(h_v^{(t)}, \bigoplus_{u\in N(v)} \varphi(h_u^{(t)}))$ with invariant $\bigoplus$ refines no finer than WL by a parallel induction, so the graph-level readout is constant on any pair WL identifies [24, 28]. □

Proposition 8 justifies the WL histogram as a principled, dependency-light stand-in for a graph-neural-network block of the descriptor.

F. The spectral-truncation kernel and the transfer policy

Proposition 9 (The spectral-truncation kernel is invariant and positive definite)**.** *Let $k(G, G')$ be the kernel of Eq. (2) on standardized features, with bandwidths $\ell_s, \ell_d > 0$. Then (i) $k(\pi G, G') = k(G, G')$ and $k(G, \pi G') = k(G, G')$ for all permutations; and (ii) k is positive definite, so for any finite set of graphs the Gram matrix $K_{ij} = k(G_i, G_j)$ is symmetric positive semidefinite.*

Proof. (i) Both feature maps are relabeling invariant, σ_r by Corollary 6 and ϕ by Theorem 7; the standardization is a fixed affine map, and k depends on G only through them, so it is invariant. (ii) Each factor is a Gaussian radial basis function on a Euclidean feature space, which is positive definite [6]. The Hadamard product of two positive-definite kernels is positive definite by the Schur product theorem, so k is positive definite. □

Corollary 10 (STK transfer is a deterministic class function)**.** *Fix a training set of (graph, optimized-schedule) pairs $\{(G_i, \theta_i)\}$ and the standardization estimated on it. For a test graph G, the STK policy outputs $\Theta(G) = \theta_{j^\star}$ with $j^\star = \arg\max_i k(G, G_i)$. Then $\Theta(\pi G) = \Theta(G)$ for every π, and Θ is a deterministic function of the isomorphism class of G given the seeded training set; ties are broken by a labeling-independent rule on training indices.*

Proof. By Proposition 9(i) the kernel values $\{k(G, G_i)\}_i$ are unchanged under relabeling of G, so the arg max index $j^\star$ and hence $\theta_{j^\star}$ are unchanged; given the fixed training set the computation has no randomness, so Θ is reproducible. □

Corollary 10 is the formal sense in which STK is a principled, invariant estimator rather than a label-dependent lookup, and is what makes the family-held-out transfer of Fig. 4 meaningful.

G. Monotone refinement, the warm-start floor, and basin locality

Proposition 11 (Monotonicity, warm-start floor, and non-equalization)**.** *Let r_t be the running-best approximation ratio after t queries of the coordinate-ascent refiner started from a schedule with ratio r_0. Then (i) r_t is monotone non-decreasing and (ii) $r_t \geq r_0$ for all t. Consequently (iii) if two policies seed the refiner in different basins of attraction of the coordinate-ascent dynamics, and the budget is exhausted before either escapes its basin, then the policy with the higher-quality basin retains its advantage: its r_t cannot be overtaken by a policy trapped below it.*

Proof. The refiner replaces the incumbent only on strict improvement, so $r_t = \max(r_{t-1}, \rho_t) \geq r_{t-1}$, which gives (i); the initial incumbent is the warm start, so $r_t \geq \cdots \geq r_0$, which gives (ii). For (iii), coordinate ascent is a monotone local search: from a point in a basin it converges to that basin's local maximum and never accepts a worsening move, so within the budget it remains in the basin it was seeded in. If policy A's basin has supremum exceeding policy B's attained value and B does not escape its basin within the budget, then $r_t^A \geq r_0^A$ stays above r_t^B. Averaging over graphs preserves (i) and (ii). □

Proposition 11 explains the depth dependence of Fig. 3: at p=1 the low-dimensional landscape is effectively unimodal over the relevant region, so all structure-aware seeds flow to the same optimum and the head start is equalized; at $p \geq 2$ the multimodal landscape traps a poorly seeded ramp in an inferior basin, so the better-seeded basin of STK is not equalized and its head start survives to the final ratio. This is the mechanism behind the growing advantage.

VI. DISCUSSION AND OUTLOOK

We have shown that the depth-one verdict on learned QAOA warm starts does not survive to larger depth. At depth one a learned, topology-conditioned warm start ties an adiabatic ramp (paired difference $+0.0000 \pm 0.0001$), reproducing the established parity. Beyond depth one the parity breaks in favor of learning: a policy that transfers a single optimized schedule from the kernel-nearest graph beats the strongest per-instance ramp by a margin that grows with depth, $+0.0103$ (final) at p=2 and $+0.0262$ at p=3, both outside their paired confidence intervals, with even larger one-shot gains. The simpler averaging learner also surpasses the ramps at depth, so the primary lesson is that topology conditioning helps once $p \geq 2$, not at p=1. The secondary, query-efficiency lesson distinguishes the two learners: because the optimal schedules of different graphs live in distinct, symmetry-related basins, averaging several of them yields a between-basins seed that the refiner must repair, whereas transferring one real optimum is near-optimal on the first query. Transfer therefore leads averaging in the low-budget and one-shot regime that matters on hardware at p=1 and p=2; by p=3 the averaging learner closes this gap, reaching final and one-shot ratios statistically indistinguishable from transfer.

The spectral-truncation kernel works because the optimal schedule at larger depth is shaped by the low-frequency structure of the instance, exactly the information carried by the smallest non-zero normalized-Laplacian eigenvalues. Truncating the spectrum to that window yields an invariant fingerprint under which a similar graph has a similar optimal schedule, so the schedule of the kernel-nearest donor is a good seed. That a finite spectral truncation suffices echoes the operator-truncation kernels of Hashimoto *et al.* [5], here with the graph Laplacian as the truncated operator and QAOA schedule transfer as the downstream task.

The framework is a contribution independent of the method. A proven relabeling-invariant descriptor and kernel, a query-counted refiner with one-shot and frontier views, a depth-one objective cross-verified by two evaluators, exact approximation ratios against a brute-force oracle, and family-held-out splits with programmatic leakage checks together form a reusable yardstick against which stronger warm-start policies can be measured. In particular it separates two questions the literature often conflates: whether a learned policy helps at all (it does, at $p \geq 2$, not at p=1), and whether the form of learning matters for efficiency (it does at p=1 and p=2, where transfer is near-optimal per query while averaging needs refinement).

Several limitations bound these conclusions and indicate natural extensions. Graphs are small ($n \leq 16$) so that exact MaxCut is tractable for ground-truth ratios; the statevector evaluator scales further, but exact oracles do not, and behavior at sizes where QAOA might offer a quantum advantage [23] is untested. The transfer targets are near-optimal schedules from an INTERP-seeded optimizer rather than certified global optima; a better labelling oracle could only raise the transfer ceiling, and the comparison with the label-free ramps is unaffected. The kernel is a fixed product of two radial basis functions with median-heuristic bandwidths and a fixed truncation order; a learned or task-weighted kernel may widen the margin. Objective values are noiseless classical evaluations without shot or device noise, the graph families are synthetic, and confidence intervals are over the held-out graphs within a single seeded run rather than over independent dataset redraws. None of these alters the present, carefully scoped claim: for depth-p QAOA MaxCut on these families and budgets, spectral-truncation-kernel transfer of angle schedules ties an adiabatic ramp at p=1 and beats it at p=2, 3 by a margin that grows with depth, and by even more on the one-shot ratio than on the final ratio at each depth. We make no quantum-hardware or asymptotic-advantage claim; the result identifies where learned QAOA warm starts help, namely beyond depth one, and supplies a transferable, theoretically grounded operator that does so.

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DATA AND CODE AVAILABILITY

This study uses no external datasets; all graphs are generated synthetically from fixed random seeds (master seed 0) by the accompanying code. The complete machine-readable results record `results/summary.json`, which contains per-policy and per-depth ratios and confidence intervals, query-budget frontier curves, per-family breakdowns, the paired advantage-versus-depth block, and run provenance, accompanies this manuscript and is the single source of truth for every number, table, and figure. The reference implementation is the CPU-only `pip`-installable package `topoqaoa`, depending on NumPy [37], SciPy [38], NetworkX [36], scikit-learn [39], and Matplotlib, with no quantum-software dependency. It contains the graph generators, the relabeling-invariant descriptors and the spectral-truncation kernel, the analytic and statevector QAOA evaluators, the INTERP-seeded schedule-labelling oracle, the budgeted refiner, all five warm-start policies, the family-held-out splitting and leakage checks, the metrics, the experiment runner, the figure and table generators, and the test suite (including the closed-form versus statevector, descriptor- and kernel-invariance, and kernel positive-definiteness checks). The reported run used Python 3.13.12, NumPy 2.4.3, SciPy 1.17.1, NetworkX 3.6.1, scikit-learn 1.8.0, and Matplotlib 3.10.8, completing in 277.6094 s at 314.1 MB peak memory on a single laptop CPU. Installation is `pip install ./submission/code`; reproduction is `topoqaoa-reproduce --config configs/full.yaml`. The package will be released in a public repository upon publication.

Appendix A: Additional numerical results

Tables III and IV give two further numerical views of the primary-depth (p=2) run. All values are generated from `results/summary.json`; no number is computed by hand.

Appendix B: A self-contained account of the constructions

This appendix gives a self-contained account of each construction the reference implementation realizes. Every symbol used in the code is defined, and each subsection is keyed to a source file under `submission/code/src/topoqaoa/`, so that the formalism and its implementation can be read together. The treatment assumes familiarity with linear algebra and elementary probability but not with QAOA or graph kernels.

1. The problem

In MaxCut one is given an undirected graph $G = (V, E)$ and seeks to split the vertices into two sets so that the number of edges crossing the split is maximal; the task is NP-hard [10]. QAOA is a quantum heuristic that prepares a parameterized state and tunes $2p$ angles to make the expected number of cut edges large. Running QAOA on hardware is expensive because every evaluation of the objective costs many circuit executions, and a warm start supplies good initial angles so that fewer evaluations are needed. The question studied here is which warm start is best as a function of depth. Topology-conditioned warm starts tie a physics-based adiabatic ramp at p=1 but beat it once $p \geq 2$, and transferring a single optimized schedule from a spectrally similar graph is a query-efficient way to do so, being near-optimal on the first evaluation.

2. Graphs and the cut function (`graph_generators.py`)

A graph $G = (V, E)$ has $n = |V|$ vertices and edge set E. A cut is an assignment $z : V \to \{+1, -1\}$. Edge (u, v) is cut if and only if $z_u \neq z_v$, that is, $(1 - z_u z_v)/2 = 1$, so the cut size is $C(z) = \sum_{(u,v)\in E}(1 - z_u z_v)/2$ and $C^\star = \max_z C(z)$. Flipping all spins gives the same cut, so we fix vertex 0 and enumerate the remaining 2^{n-1} assignments (the brute-force oracle, exact for $n \lesssim 20$; `maxcut_exact.py`). The benchmark uses six structurally distinct families (Erdős–Rényi, random regular, Barabási–Albert, Watts–Strogatz, two-dimensional grid, and stochastic block) so that generality has meaning.

3. Qubits and the QAOA state (`qaoa.py`)

A single qubit is a unit vector in $\mathbb{C}^2$; n qubits live in $\mathbb{C}^{2^n}$, and a basis state $|z\rangle$ corresponds to a spin assignment. Define the diagonal cost operator $C = \sum_{(u,v)\in E}(1 - Z_u Z_v)/2$ (so $C\,|z\rangle = C(z)\,|z\rangle$) and the transverse-field mixer $B = \sum_v X_v$. QAOA at depth p starts from $|+\rangle^{\otimes n}$ and alternates as in Eq. (1). The cost phase is diagonal, and the mixer factorizes over qubits (Lemma 2), so the $2^n \times 2^n$ matrix B is never built. `qaoa.py` implements the exact statevector evaluator at cost $O(p\, n\, 2^n)$ for arbitrary depth.

4. The depth-one closed form and cross-check (`qaoa.py`)

At p=1 the expected cut decomposes edge by edge through the analytic form of Eq. (4) [3], at cost $O(|E|)$. The code computes $\langle C\rangle$ both ways at p=1, statevector and closed form, and the test suite asserts agreement to 10^{-9} (`test_qaoa_closed_form.py`). That agreement licenses the statevector objective used at every depth.

5. Invariant descriptor and truncated spectrum (`descriptors.py`, `kernels.py`)

A learned policy must turn a graph into fixed-length vectors that do not change when the vertices are renamed, lest it exploit labels. We build two: the descriptor $\phi(G)$ from degree, motif, WL, cycle, Laplacian, and connectivity blocks (Theorem 7), and the spectral-truncation feature $\sigma_r(G)$, the r smallest non-zero eigenvalues of the normalized Laplacian $\mathcal{L} = D^{-1/2} L D^{-1/2}$ (Corollary 6). Both are invariant because they are built from spectra and multisets; the test suite verifies $\sigma_r(\pi G) = \sigma_r(G)$ and $\phi(\pi G) = \phi(G)$ to 10^{-8}.

6. Weisfeiler–Lehman colour refinement

WL colours each vertex by its degree, then repeatedly replaces a vertex's colour by a hash of its colour and the sorted multiset of neighbour colours. The histogram of colours is a relabeling-invariant fingerprint of local structure, exactly as powerful at distinguishing graphs as a

TABLE III. Per-family held-out approximation ratio at p=2. Mean depth-2 approximation ratio ±95% confidence interval ($1.96\,s/\sqrt{n}$) for each policy, with each family held out in turn. Bold marks the best mean in each row. This is the numerical companion to Fig. 4. Source: `results/summary.json` (primary-depth `by_family`).

Family	Random	Spectral	Topology	Descr. mean	STK (ours)
Barabási–Albert	0.7783 ± 0.0374	0.8383 ± 0.0149	0.8424 ± 0.0135	0.8551 ± 0.0144	$\mathbf{0.8608} \pm 0.0124$
Erdős–Rényi	0.7698 ± 0.0339	0.8528 ± 0.0128	0.8609 ± 0.0109	$\mathbf{0.8662} \pm 0.0094$	0.8655 ± 0.0100
2-D grid	0.6005 ± 0.0467	0.7577 ± 0.0066	0.7724 ± 0.0064	$\mathbf{0.7930} \pm 0.0071$	0.7897 ± 0.0083
Random regular	0.7758 ± 0.0367	0.8268 ± 0.0183	0.8359 ± 0.0171	0.8317 ± 0.0188	$\mathbf{0.8418} \pm 0.0171$
Stochastic block	0.7795 ± 0.0421	0.8548 ± 0.0196	0.8622 ± 0.0197	0.8647 ± 0.0167	$\mathbf{0.8728} \pm 0.0162$
Watts–Strogatz	0.7804 ± 0.0313	0.8607 ± 0.0128	0.8626 ± 0.0122	$\mathbf{0.8715} \pm 0.0127$	0.8714 ± 0.0130

TABLE IV. Query-budget frontier at p=2. Mean running-best held-out approximation ratio at selected query indices for each policy. The running best is monotone non-decreasing in q (Proposition 11); STK starts highest and stays above the ramps. This is the numerical companion to Fig. 3. Source: `results/summary.json` (primary-depth `frontier`).

Policy	q=1	q=2	q=4	q=8	q=16	q=28
Random	0.6057	0.6220	0.6429	0.6836	0.7210	0.7474
Spectral ramp	0.7483	0.7503	0.7537	0.8058	0.8238	0.8319
Topology ramp	0.8032	0.8112	0.8112	0.8171	0.8201	0.8394
Descriptor mean	0.8050	0.8065	0.8218	0.8302	0.8414	0.8470
STK transfer (ours)	0.8449	0.8449	0.8450	0.8452	0.8479	0.8503

basic message-passing graph neural network [28] (Proposition 8), which is why a WL histogram is a principled, dependency-light stand-in for such a block.

7. Why transfer is query-efficient (`kernels.py`, `baselines.py`)

Optimal QAOA schedules are not unique: time-reversal and mixer/cost periodicities place equally good schedules in several symmetry-related basins. If a learned policy averages the optimized schedules of several training graphs, as a regressor does, the average can fall between basins and seed the refiner in a region it must then climb out of; the policy can still reach a good final ratio once enough budget is spent, but its one-shot quality suffers. Transferring a single optimized schedule from one similar graph is by construction a real optimum in one basin, so it is near-optimal immediately, the property that matters when each query is expensive. The spectral-truncation kernel $k(G, G')$ of Eq. (2), a product of an RBF on the truncated spectrum and an RBF on the descriptor (Proposition 9), chooses the donor: the training graph whose low-frequency Laplacian structure is most similar, hence whose optimal schedule transfers best. The `STKPolicy` returns the kernel-nearest donor's schedule; a kernel-ridge variant (averaging) is shipped as an ablation. Empirically both learned policies beat the adiabatic ramps at $p \geq 2$; transfer attains the best one-shot ratio at p=1 and p=2 and remains far above the ramps at p=3, where the averaging learner draws level.

8. Schedule-labelling oracle and the refiner (`qaoa.py`, `env.py`)

Transfer targets are near-optimal schedules found by INTERP-seeded optimization [4]: optimize p=1 on the exact grid, grow to depth p by interpolating the schedule, and polish with Nelder–Mead, on training graphs only. At test time, hardware cost scales with the number of objective evaluations, so the refiner counts them: coordinate ascent over all $2p$ angles from the warm start, with initial step 0.3, halving on stalls, and halting at the budget. The query-budget frontier is the mean running-best ratio versus query index; its first point is the one-shot ratio, a direct measure of warm-start quality.

9. Splits, leakage, metrics, and significance (`splits.py`, `metrics.py`, `runner.py`)

Transfer is tested family-held-out: to evaluate a family F, the learned policies are fit only on the other five families. A fingerprint (family, n, $|E|$, hashed degree sequence) checks that no test graph appears in training (`leakage_clean`= true). The primary metric is the exact approximation ratio $\langle C \rangle / C^\star \in [0, 1]$; advantages over baselines are paired across graphs with a 95% confidence interval $1.96\,s/\sqrt{n}$ on the paired difference. `runner.py` ties everything together and writes `results/summary.json`, the single source of truth from which every table, figure, and macro is generated, with nothing entered by hand.

10. Reproducing the run

From `submission/code/` (after `pip install .`, or with `PYTHONPATH=src`): `make test` runs the cross-checks (10^{-9}, 10^{-8}, and kernel positive-definiteness); `python3 scripts/run.py --config configs/full.yaml --out results` produces the results record; `python3 scripts/make_tables.py` writes the macros and tables; and `python3 scripts/make_figures.py` writes the four figure PDFs. The reported run uses 84 graphs across 6 families, depths $1, 2, 3$, budget 28, and seed 0, for which STK ties the ramp at p=1 and beats it by +0.0262 (final) at p=3.

11. Scope of the result

At depth one a learned topology warm start and a one-line adiabatic ramp reach the same approximation ratio, because both encode the same single-scale signal. As depth grows the warm start must specify a whole schedule, and topology-conditioned warm starts beat the ramp by a margin that grows with depth. Transferring one optimized schedule from the spectrally nearest graph is near-optimal on the first query: at p=1 and p=2 this gives transfer the best one-shot ratio, while averaging-based learning must still repair a between-basins seed, and by p=3 the two draw level. The result is a transferable, theoretically grounded warm-start operator and a controlled, cross-verified, leakage-checked benchmark that isolates where learned QAOA warm starts help. No quantum-hardware or asymptotic-advantage claim is made; the depth is at most three, the sizes are $n \leq 16$, and the graphs are synthetic.

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